

Text 2: Automation and Optimization of Telemetry Data Collection on Mobile

Integrating telemetry modules into a mobile application opens new possibilities for monitoring, analysis, and optimization of usage. Whether for connected health applications, industrial maintenance, or fleet tracking, the ability to collect real-time data via sensors (BLE, NFC, GPS, accelerometer) is a strategic asset. Automating data collection requires setting up reliable mechanisms for synchronization, handling network interruptions, and conserving energy.

To maximize reliability, it is crucial to implement error recovery logic: in case of connection loss, data is stored locally (for example, using Room or Realm), then synchronized with the server once connectivity is restored. Using WorkManager, combined with contextual triggers (battery level, WiFi only), optimizes the scheduling of collection and sending tasks. Data compression, through algorithms such as Gzip, helps reduce bandwidth consumption and speed up transfers.

Another fundamental aspect is securing communications: encrypting sensitive data, strong authentication, and access control management are essential, especially in sensitive contexts like healthcare or automotive. Implementing robust and well-documented REST APIs (Swagger/OpenAPI) simplifies integration with the company's information systems.

Finally, leveraging the collected data involves creating analytical dashboards to visualize trends, identify anomalies, and optimize business processes. Artificial intelligence applied to these large data volumes provides new levers for predictive optimization: preventive maintenance, user experience personalization, risk behavior detection. Performance monitoring must be continuous, with proactive alerts in case of drift or critical incidents.